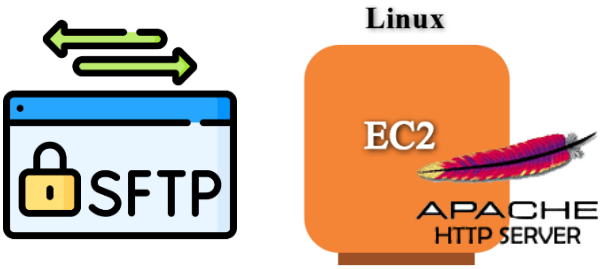
AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

7.Configure SFTP to Linux Web Server in AWS



Santhu Mohammed M
CSE-Artificial Intelligence & Machine Learning,

Sri Eshwar College of Engineering,

Coimbatore.

Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

Use Cases for EC2 Linux Instances:

- Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.
- Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.
- Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.
- Development and Testing: Create development environments with linux OS to test php,MySql or other Linux-based applications.

SFTP- Secure File Transfer Protocol

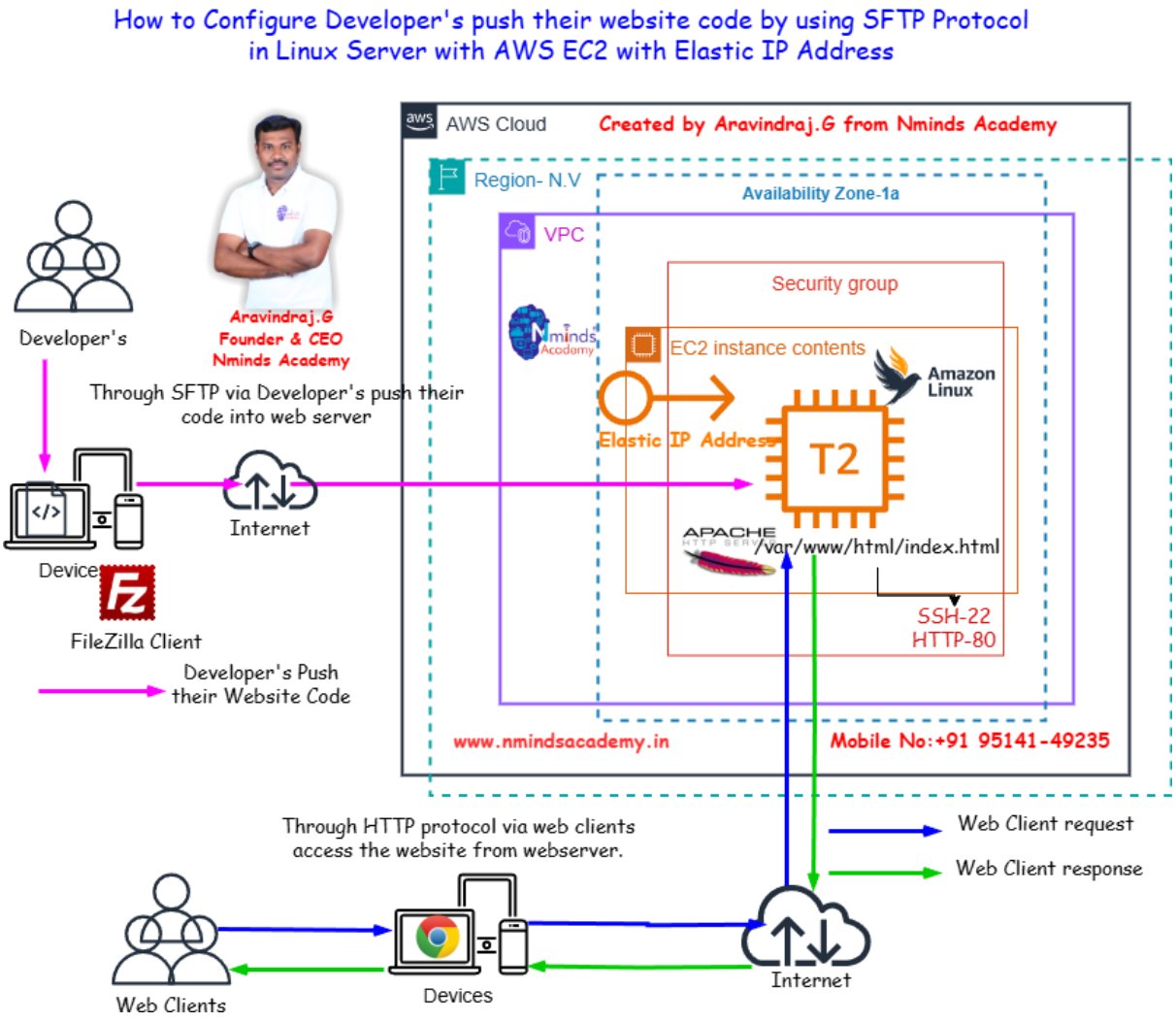
SFTP (Secure File Transfer Protocol) is a secure network protocol that allows users to transfer files over a secure, encrypted connection. It is a more secure alternative to FTP (File Transfer Protocol) and is commonly used to upload, download, and manage files on remote servers.

Here are the key uses and advantages of SFTP:

1. File Transfer: SFTP allows secure transfer of files between a local and a remote computer over a network. It ensures that the data is encrypted during transmission, protecting it from eavesdropping or tampering.
2. File Management: Besides transferring files, SFTP also allows users to perform file management tasks such as renaming, deleting, and moving files on the remote server.
3. Secure Authentication: SFTP uses SSH (Secure Shell) for authentication, which provides a stronger level of security compared to FTP's traditional username/password system. It often supports public key authentication, adding an extra layer of security.
4. Encryption: The main advantage of SFTP over FTP is encryption. All data sent over SFTP is encrypted, ensuring that sensitive information (like passwords and files) is protected during transmission.

5. Firewall-Friendly: SFTP operates over a single port (usually port 22), making it easier to use behind firewalls and NAT (Network Address Translation) devices compared to FTP, which uses multiple ports.

Topology



Used Services in AWS

Used Services in Windows Server

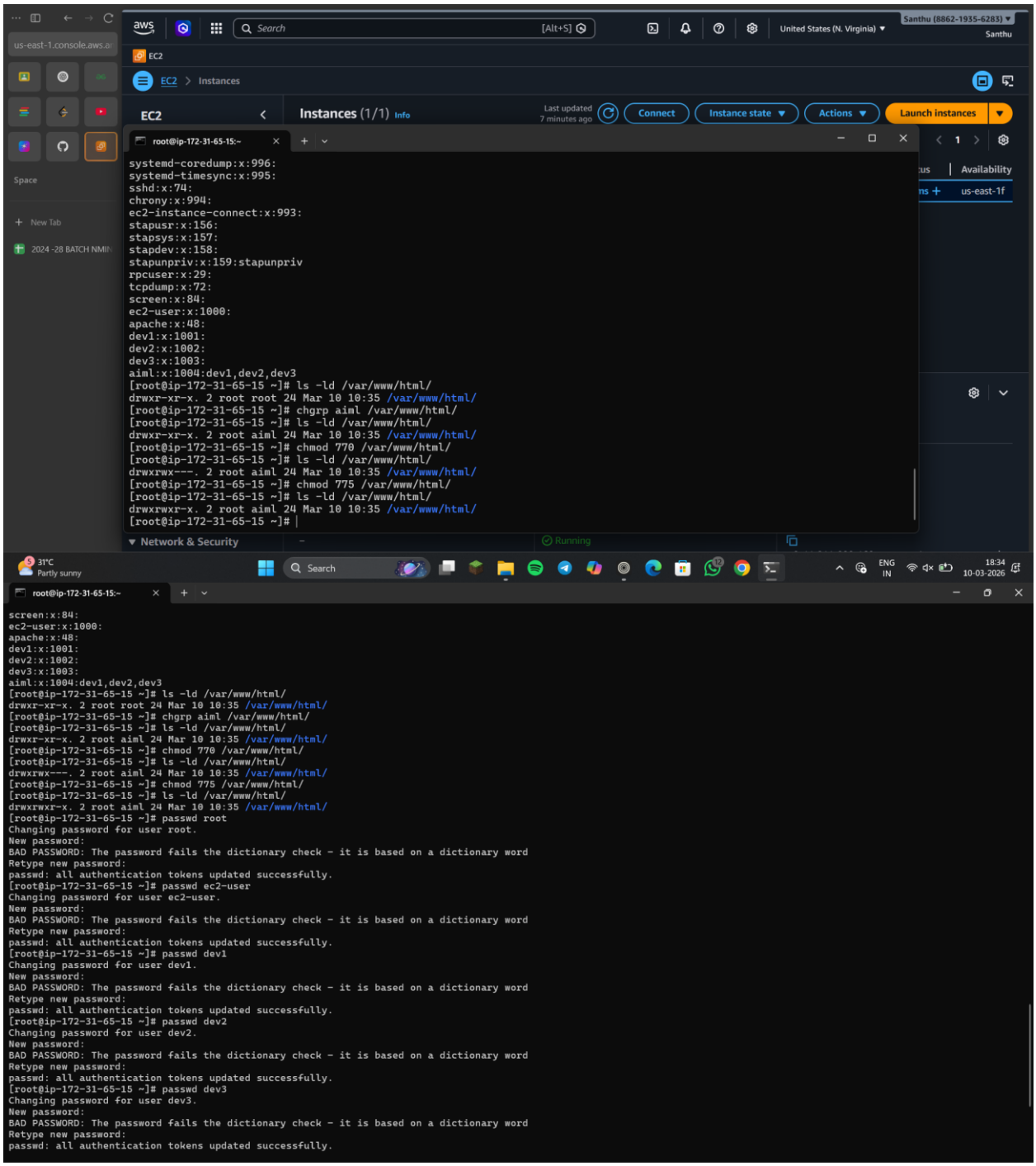
Amazon EC2

APACHE HTTP SERVER

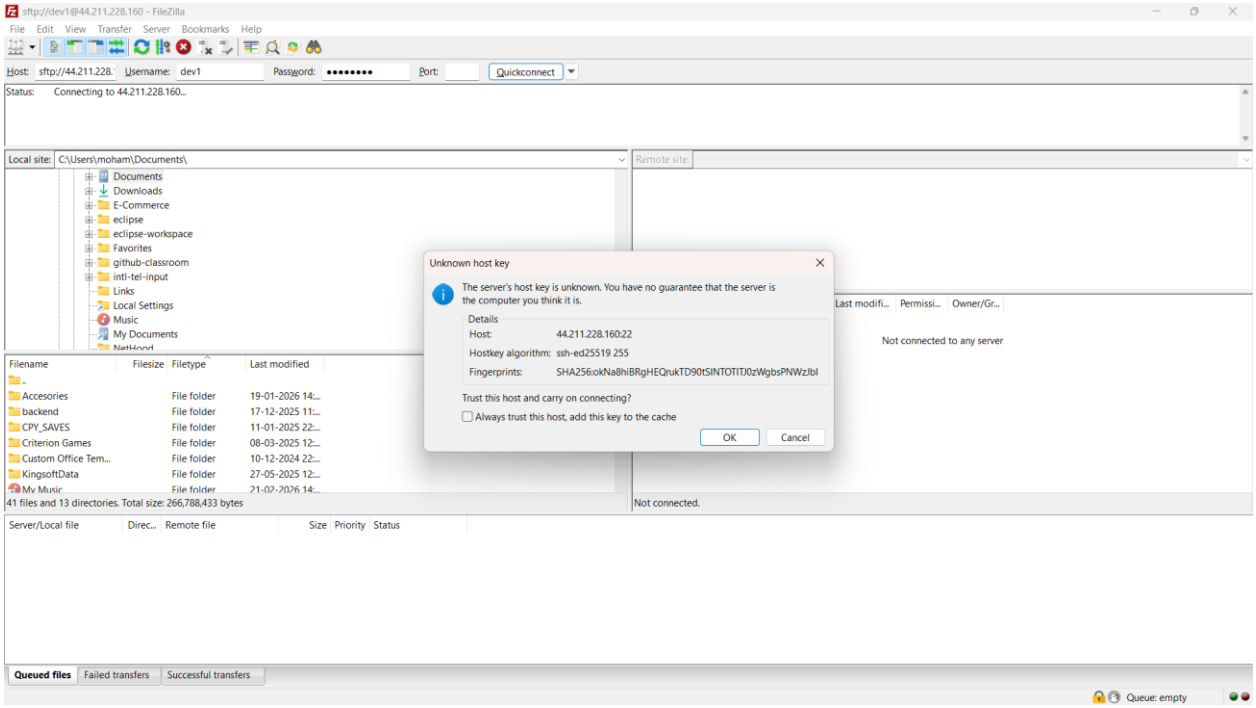
Apache Webserver

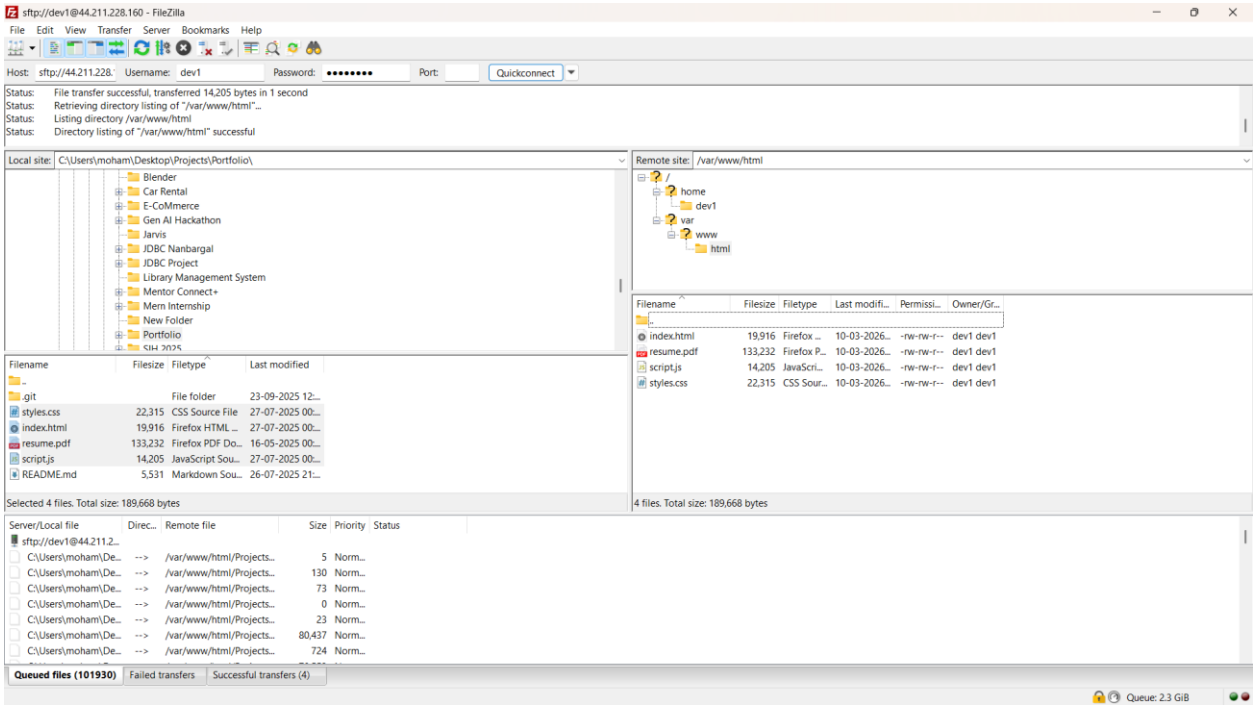
Amazon Elastic IP Address

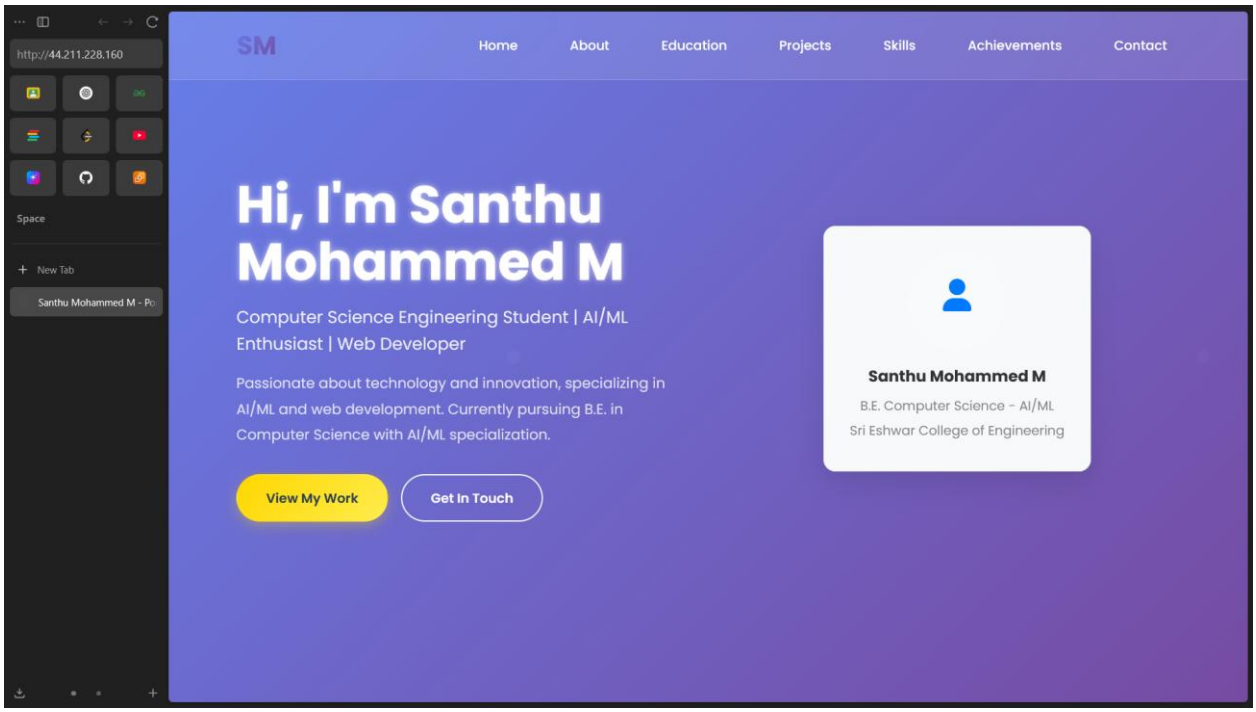
SFTP



```
root@ip-172-31-65-15:~

#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile      .ssh/authorized_keys

#AuthorizedPrincipalsFile none


# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no

# Change to no to disable s/key passwords
#KbdInteractiveAuthentication yes

"/etc/ssh/sshd_config" 133L, 3854B                                    65,23         30%
```




sftp://dev1@44.211.228.160 - FileZilla
File Edit View Transfer Server Bookmarks Help
Host: sftp://44.211.228. Username: dev1 Password: •••••••• Port: Quickconnect
Status: File transfer successful, transferred 14,205 bytes in 1 second
Status: Retrieving directory listing of "/var/www/html"...
Status: Listing directory /var/www/html
Status: Directory listing of "/var/www/html" successful
Local site: C:\Users\moham\Desktop\Projects\Portfolio\
Blender
Car Rental
E-CoMmerce
Gen AI Hackathon
Jarvis
JDBC Nanbargal
JDBC Project
Library Management System
Mentor Connect+
Mern Internship
New Folder
Portfolio
Remote site: /var/www/html
home
dev1
var
www
html
Filename Filesize Filetype Last modified
.git File folder 23-09-2025 12:...
styles.css 22,315 CSS Source File 27-07-2025 00:...
index.html 19,916 Firefox HTML ... 27-07-2025 00:...
resume.pdf 133,232 Firefox PDF Do... 16-05-2025 00:...
script.js 14,205 JavaScript Sou... 27-07-2025 00:...
README.md 5,531 Markdown Sou... 26-07-2025 21:...
Selected 4 files. Total size: 189,668 bytes
Filename Filesize Filetype Last modifi... Permissi... Owner/Gr...
index.html 19,916 Firefox ... 10-03-2026... -rw-rw-r-- dev1 dev1
resume.pdf 133,232 Firefox P... 10-03-2026... -rw-rw-r-- dev1 dev1
script.js 14,205 JavaScri... 10-03-2026... -rw-rw-r-- dev1 dev1
styles.css 22,315 CSS Sour... 10-03-2026... -rw-rw-r-- dev1 dev1
4 files. Total size: 189,668 bytes
Server/Local file Direc... Remote file Size Priority Status
Queued files (101930) Failed transfers Successful transfers (4)
Queue: 2.3 GiB


http://44.211.228.160
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New Tab
Santhu Mohammed M - Po
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Hi, I'm Santhu Mohammed M
Computer Science Engineering Student | AI/ML Enthusiast | Web Developer
Passionate about technology and innovation, specializing in AI/ML and web development. Currently pursuing B.E. in Computer Science with AI/ML specialization.
View My Work
Get In Touch
Santhu Mohammed M
B.E. Computer Science - AI/ML
Sri Eshwar College of Engineering